

Aviation Flight Security Assurance Evaluation System Based on Panboolean Algebra

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Abstract—The quality of aviation security communications directly affects the quality of flight and even the safety of flight. An assessment of the communications security of each flight provides a powerful reference for follow-up support. According to the reality of aviation security protection, taking the protection of communication in a flight schedule as the research object, the main factors affecting the protection are fully considered, A system evaluation mathematics evaluation model was established by means of the Pan-Boolean algebra method with more intuitive laws in the principle of automatic control. The example shows that the evaluation made by the method of the Pan-Boolean algebra can provide the commander with objective and fair data evaluation, and it also provides data support for the next step of communication security personnel, so as to better ensure the foundation of flight.

Keywords—aviation flight; communication security; systematic evaluation; panboolean algebra

I. INTRODUCTION

In order to adapt to the modern international environment and respond to the various uncertain challenges in the future, we need to make corresponding assessments of the existing communication support capabilities. As an aviation force's communications support capability, it is mainly determined by three major factors: how many people are protected, how many times equipment fails, and whether management is scientifically regulated. Reasonable force distribution, low equipment failure rate, and scientific and standardized management are the criteria for evaluating whether the guarantee is high quality and high efficiency. There are many evaluation methods at present, such as expert delphi, ideal point method, Analytic Hierarchy Process (AHP), and Analytic Network Process (ANP). According to research, the delphi method relied more on the experience of experts. After each flight, the commander commented on flight guarantees by his impressions and comments. Analytic Hierarchy Process (AHP), if the selected factors are unreasonable, its meaning is confusing, or the relationship between factors is not correct, will reduce the quality of the AHP method, and even lead to failure of the AHP method[1]. However, the traditional ANP assessment method requires that each factor has a relationship of interaction and mutual influence. The modeling process is relatively complex and cumbersome, resulting in an increase in the workload of people and higher costs. The Pan-Boolean algebra control originated in the early 80s. It differs from the Boolean algebra in that the

classification law replaces the complementarity law and the corresponding undefined changes[2]. After several decades of development, the control theory of Pan-Bauer has become more and more mature. This article is a mathematical model built with the Pan-Bauer algebra method. There is no necessary connection between the factors that affect evaluation, and they are also independent and pass through each other. The logic control model established by the Pan-Boolean algebraic method can be converted into the control model of ordinary algebra[3], which is more in line with human logic thinking, simple and practical, and easy to understand.

II. AVIATION AIRPORT COMMUNICATION GUARANTEE EVALUATION SYSTEM BASED ON PANBOOLEAN ALGEBRA

A. Air Force Airport Communication Guarantee

The communications security of aviation airfields mainly includes ground communications, navigation, and command automation assurance tasks[2]. The main tasks are: responsible for ground and ground communications at all levels, establishing and maintaining communication liaison, organizing and implementing navigation guarantees, establishing and managing command automation systems, taking charge of communication electronic defense and communication facilities protection, and organizing battlefield radio management.

One of the problems that plagues commanders, airlines, and support personnel at present is how to objectively and impartially make a reasonable assessment of security work after the flight is over. While the commander has one flight schedule, he is basically on the outside field tower, and communication security is a huge integrated task. So how do you evaluate the safety of a flight? The first is to participate in the protection of personnel, the second is to participate in the protection of communications equipment, third, the management of scientific and standardized, such as the completion of the protection card. These three factors constitute the three main factors of Pan-Bauer algebra evaluation. Using Pan-Bauer algebra methods, a scientific and intuitive flight communications security evaluation is carried out on these three factors.

B. Evaluation System

Evaluation System: The rapid development of the economy has led to increasingly fierce competition in all walks of life. The requirements for quality standards for customer service are also increasing. How to accurately judge the quality of service,

thereby improving the service quality of the unit, and improving the enthusiasm of the service staff is an urgent issue facing the unit's leaders. In order to improve service quality of service personnel and improve service customer satisfaction, an evaluation system has been developed. The development of the evaluation system can help the customer to evaluate the service quality of the service personnel after the completion of the business, to facilitate the management of the service quality of the service personnel, to promote the unit to improve services, and improve customer satisfaction.

In the final analysis, communication security is centered on the improvement of the "flying support capability". Its purpose is to truly realize information security, automation, integration, and scientification of flight support, so as to achieve more accurate, timely, and safe flight assurance tasks^[3]. According to the actual situation of the current aviation force, the evaluation of communication security is subjective and random. The basic commander of flying commentary basically relies on feelings and impressions, and flight protection personnel cannot recognize their own level of protection and deficiencies, resulting in passive work. Therefore, the communication assurance evaluation system can reflect these factors with data and has important practical significance in flight guarantee.

C. Pan Boolean Algebra Theory

The Pan Boolean algebra system is to set up a mathematical model of a specific and complex system, and form a set of axiom systems that can make the results of the modeling simplest, thus making the information processing process simple and effective. The mathematical model of the Pan-Bauberg algebra is based on the model of output and input causality, that is, the logical model. The basic idea of the Pan Boolean algebra is to make English letters in uppercase X_1 ,

X_2 , X_3 ...to expresses variables in the inspection system.

The total number of all states in which any of the variables X_i may appear is referred to as the "state number" corresponding to X_i , and expressed as ($i=1,2,\dots$), also use English letters with subscripts and subscripts, also use the English letters with the subscripts x_i^j ($i=1, 2, \dots j=1, 2, n_i$) and subscripts to represent the j -th state corresponding to X_i . That is, when the j -th state occurs, it takes a value of 1, and when the j -th state does not occur, it takes a value of 0. In a multi-state system, each factor X_i is on a certain occasion, take and only take one of the states n_i , this is determined by the rules of division of the concept, the various states of division are incompatible with each other, and the logical sum of the divided states must exhaust the divided variables. This requires that it meet the following two conditions at the same time: ① $x_i^1 + x_i^2 + \dots + x_i^j = 1$; ② for any j and k , if $1 \leq j < k \leq n_i$, then $x_i^j \cdot x_i^k = 0$. Except that these two conditions are different from the complement laws in Boolean algebra, the rest are such as the combination law, the exchange law, the

distribution law, and the 0-1 law, both are similar to the related laws in Boolean algebra, so they are called Pan-Boolean algebras.

Pan-Bauer algebra is applied to multi-state systems and can reflect the real logic of things. It provides a relatively simple set of mathematical methods, strictly distinguishes And each state value of data mathematically, each state value has a unique, pan-Boolean expression with no missing values nor duplicate values, which can describe the system's decision rules, can apply its axiom system to simplify the decision rules accordingly, and get the simplest decision rule pattern^[4].

III. MATHEMATICAL MODEL ESTABLISHMENT AND EXAMPLES

When a subsystem in large-scale network fails due to the impact of internal or external disturbances, the effects on other subsystems can be analyzed by utilizing the survivability association function among multiple subsystems. For the large-scale network, the degree of survivability transmission and expansion among subsystems is related with survivability association function among them. Typically, the variables of survivability association function contains the similarity degree of survivability status among subsystems in large-scale network. Below, we investigate and propose the survivability association function of large-scale network.

The selection and evaluation of the evaluation system in accordance with actual evaluation indicators is an important part of communication security to form combat effectiveness. On the one hand, it can measure the strength of current communication support capabilities, and it can also reflect the gap between the level of protection and the expectations of commanders. At the scene of flight protection, the ability to comment on communications security is actually an overall assessment of the operational status of flight communications support personnel and communications equipment^[5]. Combining the practical and extensive work experience of current aviation force communications, by classifying, organizing, and analyzing, and being able to make it clear, several factors have been initially proposed to determine the ability to support: Number of participants X_1 , Registered quality X_2 , And equipment failures X_3 . According to these three factors, it is determined that this flight communication guarantee is excellent, good, qualified, and unqualified. And determine the following criteria: fewer protection personnel, more standardized registration, less equipment failure, the better the protection, and vice versa, the worse.

According to the axiom system of Panboolean algebra, we can establish the system's pan-Boolean control model. The number of states for the three factors is: $n_1=3$, $n_2=2$, $n_3=3$, That is, x_1^1 (there are many people participating in the protection). x_1^2 (The number of participants

participating in the protection is normal), x_1^3 (The number of people participating in the protection is low), x_2^1 (Security card is not standardized), x_2^2 (safety card fill in specification), x_3^1 (equipment failures occur frequently), x_3^2 (equipment failures are normal), x_3^3 (equipment failures are normal). In addition, the output factor is represented by F, the number of states is 4, which is f_1 (unqualified), f_2 (qualified), f_3 (good), f_4 (Excellent).

The system logic diagram is as follows:

$X_3 \backslash X_1$		X_2		x_1^1	x_2^2
		x_1^1	x_2^2		
x_3^1	x_1^1	$x_1^1 x_2^1 x_3^1$	$x_1^1 x_2^2 x_3^1$	$x_1^1 x_2^1 x_3^1$	$x_1^1 x_2^2 x_3^1$
	x_1^2	$x_1^2 x_2^1 x_3^1$	$x_1^2 x_2^2 x_3^1$		
	x_1^3	$x_1^3 x_2^1 x_3^1$	$x_1^3 x_2^2 x_3^1$		
x_3^2	x_1^1	$x_1^1 x_2^1 x_3^2$	$x_1^1 x_2^2 x_3^2$	$x_1^1 x_2^1 x_3^2$	$x_1^1 x_2^2 x_3^2$
	x_1^2	$x_1^2 x_2^1 x_3^2$	$x_1^2 x_2^2 x_3^2$		
	x_1^3	$x_1^3 x_2^1 x_3^2$	$x_1^3 x_2^2 x_3^2$		
x_3^3	x_1^1	$x_1^1 x_2^1 x_3^3$	$x_1^1 x_2^2 x_3^3$	$x_1^1 x_2^1 x_3^3$	$x_1^1 x_2^2 x_3^3$
	x_1^2	$x_1^2 x_2^1 x_3^3$	$x_1^2 x_2^2 x_3^3$		
	x_1^3	$x_1^3 x_2^1 x_3^3$	$x_1^3 x_2^2 x_3^3$		

According to the control rules, There are $3 \times 2 \times 3 = 18$ actual conditions as follows, (1) When the input is $x_1^1 x_2^1 x_3^1$, the output is y_1 , (2) When the input is $x_1^1 x_2^2 x_3^1$, the output is y_1 , (3) When the input is $x_1^2 x_2^1 x_3^1$, the output is y_1 , (4) When the input is $x_1^2 x_2^2 x_3^1$, the output is y_1 , (5) When the input is $x_1^3 x_2^1 x_3^1$, the output is y_1 , (6) When the input is $x_1^3 x_2^2 x_3^1$, the output is y_1 , (7) When the input is $x_1^1 x_2^1 x_3^2$, the output is y_1 , (8) When the input is $x_1^1 x_2^2 x_3^2$, the output is y_2 , (9) When the input is $x_1^2 x_2^1 x_3^2$, the output is y_2 , (10) When the input is $x_1^2 x_2^2 x_3^2$, the output is y_2 , (11) When the input is $x_1^3 x_2^1 x_3^2$, the output is y_2 , (12) When the input is $x_1^3 x_2^2 x_3^2$, the output is y_2 , (13) When the input is $x_1^1 x_2^1 x_3^3$, the output is y_2 , (14) When the input is

$x_1^3 x_2^1 x_3^3$, the output is y_2 , (15) When the input is $x_1^2 x_2^2 x_3^3$, the output is y_2 , (16) When the input is $x_1^3 x_2^2 x_3^3$, the output is y_3 , (16) When the input is $x_1^2 x_2^2 x_3^3$, the output is y_2 , (17) When the input is $x_1^2 x_2^2 x_3^3$, the output is y_3 , (18) When the input is $x_1^3 x_2^2 x_3^3$, the output is y_4 .

According to the logical relationship can be written control expression:

$y_1 = x_1^1 x_2^1 x_3^1 + x_1^2 x_2^1 x_3^1 + x_1^3 x_2^1 x_3^1 + x_1^1 x_2^2 x_3^1 + x_1^2 x_2^2 x_3^1 + x_1^3 x_2^2 x_3^1$, These 7 states represent y_1 .

$y_2 = x_1^1 x_2^2 x_3^2 + x_1^2 x_2^2 x_3^2 + x_1^3 x_2^2 x_3^2 + x_1^1 x_2^1 x_3^3 + x_1^2 x_2^1 x_3^3 + x_1^3 x_2^1 x_3^3$, These 7 states represent y_2 .

$y_3 = x_1^2 x_2^2 x_3^3 + x_1^3 x_2^2 x_3^3 + x_1^1 x_2^2 x_3^3$, These 3 states represent y_3 .

$y_4 = x_1^3 x_2^2 x_3^3$, These 1 states represent y_4 .

Example: If an ordinary flight is involved in communication security protection for 32 persons, there are 2 failures during the flight, and the security card is not filled in. According to Panboolean algebraic model and system standard library, we can see: Factor 1 is x_1^2 , Factor 2 is x_2^1 , Factor 3 is x_3^2 , which can be judged as unqualified.

Confirmation of the person X_1 participating in the protection: Each flight guarantee is signed and confirmed by the protection personnel, and the actual number of signatures shall prevail; Whether the security card is fill X_2 ed or not is determined: compare with the degree of standardization of the security card; Determine the number of equipment failures X_3 : According to the number of equipment failures, the number of equipment failures is determined by the number of equipment alarms. From this it can be seen that for each flight communications communication guarantee, these three factors are independent and can be clearly judged, and can be mathematically described by Panboolean algebra to establish a parsable mathematical model that can ultimately serve as commanders and Safeguards provide a clear assessment of safeguards.

IV. CONCLUSIONS

In this paper, we clarify the concept of large-scale network survivability from the perspective of complex system, analyze the survivability association of large-scale network based upon the association characteristics of complex system, and establish

large-scale network survivability association structure model and analyze its natures. Then, we define the survivability association function of large-scale network based upon set pair analysis theory, which is used to depict the degree of survivability association among subsystems in large-scale network. Finally, we validate the effectiveness of the proposed model through case study. The future research work includes the dynamics of survivability association in large-scale network and the application of survivability association model to large-scale network analysis.

Flight communications security is a complex system work. the author participated in the flight communications security work for nearly ten years, different levels and types of flights, the number of communications guarantees and equipment failure references are different. This paper presents a theoretical method for judgment. After the actual application, it is necessary to make corresponding adjustments along with the changes in the technical level of

personnel and the replacement of equipment, and do a great deal of data and data collection. This is the focus of future work.

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